

Applied Machine Learning: From Fundamentals to Sustainable Applications

By

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Total number of Grades: 6

Exam outcome: Pass or Fail

Objectives:

After completing the course, the student should be able to

1. Explain basic concepts in machine learning and deep learning
2. Apply common algorithms in supervised and unsupervised learning
3. Understand the basics of physics-informed AI
4. Implement and train neural networks, including convolutional networks (CNNs)
5. Analyze applications of machine learning in real-world scenarios with a focus on sustainability, energy, industry, agriculture, and health.

- **Week 1**

Introduction to Machine learning , Google Colab and Basic data processing steps

- **Week 2**

Linear regression and Logistic regression, Lab session

- **Week 3**

Neural Networks and back propagation, Neural Networks practical

- **Week 4**

PINNs

- **Week 5**

Other classifiers and Clustering

- **Week 6**

Introduction to CNNs, CNNs (demo to Image classification)

- **Week 7 & 8**

Sunil & Nusyba (on demand, no fixed hours)

Project doubt clarification and Student meetings as needed

Course Requirements & Evaluation

- **Quizzes (3 total as homework / Assignment)**
 - Unlimited attempts
 - Designed to help you understand the basics
 - Must be completed before your presentation
- **Assignments (2 total)**
- **Final Project**
 - **Project Report + Presentation**
 - Presentation: **3 minutes**, followed by **2 minutes of Q&A**

Project Guidelines

- The goal of the project report and presentation is not only evaluation, but also to help other students learn about:
 - Different applications of machine learning
 - Various approaches to solving real-world problems
- You are encouraged to:
 - Choose a problem and dataset related to your own field or area of expertise
 - Alternatively, you may select from the provided project options
 - Seek guidance from instructors when needed
- Use of AI Tools
- You are encouraged to use AI tools (e.g., ChatGPT, Copilot) for:
 - Coding assistance
 - Improving writing
- Note:
 - You must clearly mention where AI tools were used
 - You are responsible for verifying and validating all AI-generated content
 - The final report must reflect your understanding and accuracy

Team Size

- A maximum of two students per project

Presentation Date: June 17